

Unit 1, Lesson 4: Operations with Integers

Benchmarks

MA.6.NSO.4.1 Apply and extend previous understandings of operations with whole numbers to add and subtract integers with procedural fluency.

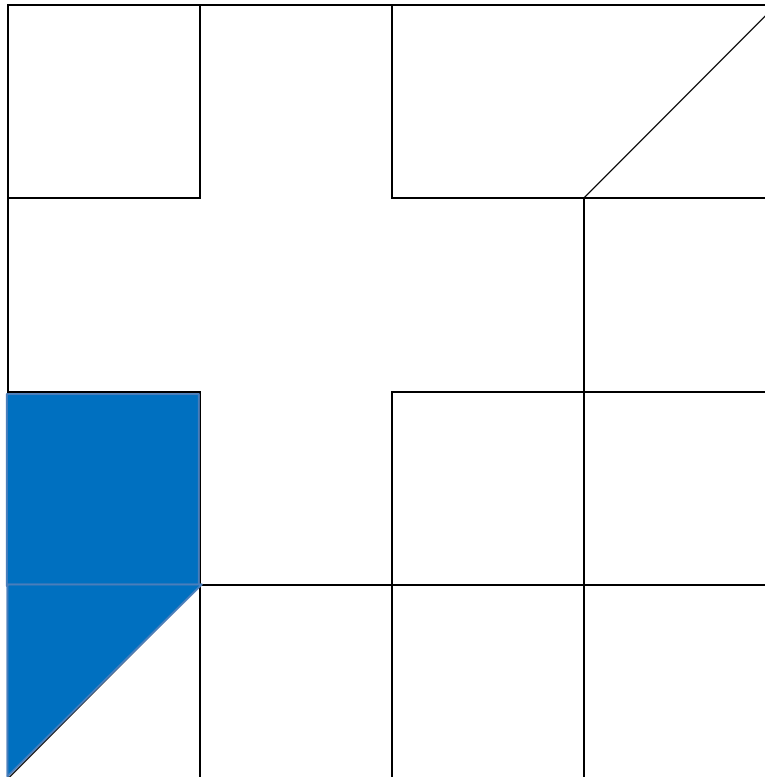
MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.

Learning Target

- ☐ I can add, subtract, multiply, and divide integers.

1.4.1 Warm-Up: Critical Thinking

1. A design is shown, where one part of the design is shaded.



- a. What fraction of the diagram is shaded?
- b. Explain how you determined your answer.

1.4.2 Activity: Card Sort

1. Work with your partner or group to match each expression with the integer chip model and number line model that represents the expression. Record your answers in the table. Then, use the models to determine the value of each expression.

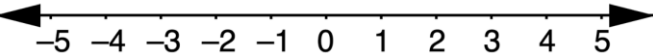
Mathematical Expression	Letter of Integer Chip Model	Letter of Number Line Model	Value
$4 + (-7)$			
$-3 + (-2)$			
$7 - 10$			
$-5 - 3$			
$-8 - (-2)$			
$4 - (-6)$			
$4 + (-5)$			
$-2 + (-8)$			

2. Explain the method you used to match the cards together.

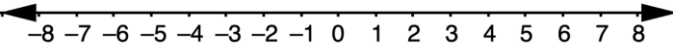
1.4.3 Guided Practice: Adding and Subtracting Integers

1. Model adding or subtracting the two integers by using the integer chip model and the number line model.

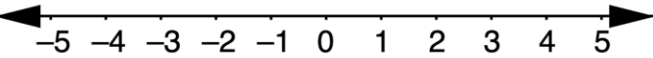
a. $5 + (-2) =$ _____

Integer Chip Model	Number Line Model
	

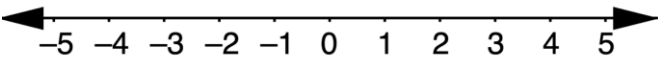
b. $-4 - 3 =$ _____

Integer Chip Model	Number Line Model
	

c. $(-1) + (-2) =$ _____

Integer Chip Model	Number Line Model
	

d. $-1 - (-4) =$ _____

Integer Chip Model	Number Line Model
	

2. Complete the statements.

When adding integers of different signs using the integer chip model, the smaller number of chips will combine with the same number of chips of the opposite sign

to form

☐ the sum.

☐ zero

The remaining chips show

☐ the sum.

☐ zero

The sum has the sign of the

☐ larger

☐ smaller

 number of chips.

3. Complete the table.

Expression	Number of Zero Pairs Created	Sign of Sum	Sum
$(-9) + 12$	9	+	3
$13 + -7$			
$(-10) + 5$			
$7 + (-7)$			

1.4.4 Bring It Together: Adding and Subtracting Integers

1. Complete the statements.

a. When adding integers of the same sign,

- ☐ add
- ☐ subtract

the absolute values of the integers. The sum will have the

- ☐ same
- ☐ opposite

sign as the integers being added.

b. When adding integers of different signs,

- ☐ add
- ☐ subtract

the absolute values of the integers. The sum will have the same sign as the

integer with the

- ☐ larger
- ☐ smaller

absolute value.

c. When subtracting integers, create an equivalent expression by changing the

expression to

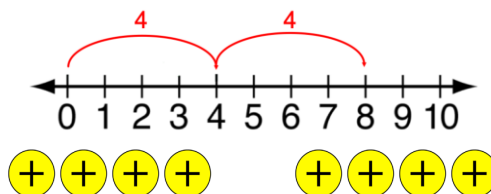
- ☐ addition
- ☐ multiplication

of

- ☐ a zero pair.
- ☐ the additive inverse.

1.4.5 Guided Practice: Multiplying and Dividing Integers

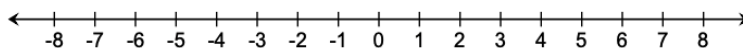
The expression 2×4 means 2 groups with 4 units in each group. This can be represented on the number line or using integer chips as shown.



1. The expression $2 \times (-4)$ is given.

a. Create 2 groups of -4 using the integer chip model.

b. Model 2 groups of 4 units to the left of 0.



c. Complete the equation.

$$2 \times (-4) = \underline{\hspace{2cm}}$$

d. Explain how the expressions $(-2) \times 4$ and $2 \times (-4)$ differ.

2. Complete the visual rule for multiplication of positive and negative numbers.

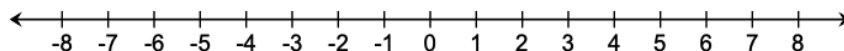
Visual Model of Multiplication	Sign of Product
$\times$	
$\times$	

3. Complete the statement.

The expression $(-2)(-4)$ indicates that groups
of should be **removed**.

4. Consider how to represent the expression $(-2) \times (-4)$.
- a. Create an integer chip model to represent this expression.

- b. Model the expression on the number line.



- c. Complete the statement.

$$(-2) \times (-4) = \underline{\hspace{2cm}}$$

5. Complete the visual rule for the multiplication of two negative numbers.

Visual Model of Multiplication	Sign of Product
 $\times$ 	

6. Multiplication and division have an inverse relationship. Use this relationship to complete the table.

Given Equation	Can Be Thought of As ...	Related Equation
$3 \times \underline{\hspace{1cm}} = 21$		$21 \div 3 = \underline{\hspace{1cm}}$
$-8 \times \underline{\hspace{1cm}} = 56$		$56 \div (-8) = \underline{\hspace{1cm}}$
$20 \div \underline{\hspace{1cm}} = -5$		$(-5)(\underline{\hspace{1cm}}) = 20$
$-\left(\frac{36}{4}\right) = \underline{\hspace{1cm}}$		$\underline{\hspace{1cm}} \times 4 = -36$

7. How are the rules for multiplication of integers related to the rules for division of integers?

1.4.6 Try It: Integer Operations

1. Evaluate each expression.

a. $34 + (-21)$

b. $-67 + 40$

c. $-333 + (-509)$

d. $700 + (-814)$

e. $632 - (-108)$

f. $942 - 621$

g. $-78 - (-324)$

h. $105 - (-75)$

i. $-5 \times (-6)$

j. $-22 \div (-11)$

k. $-24 \div 3$

l. $9 \times (-4)$

1.4.7 Wrap-Up: Reflection

1. In this lesson, multiple concepts about operations with integers were reviewed. Rate yourself on the following skills based on your level of confidence.

Statement	Rating 1 – I struggle with this. 2 – I have some skills. 3 – I am almost there. 4 – I have got this!
I can add integers.	
I can subtract integers.	
I can multiply integers.	
I can divide integers.	